

Lab no 1

Process System Calls

Objectives:

- Overview of kernel mode and user mode.
- Implementation of DOS-command (Internal and External)

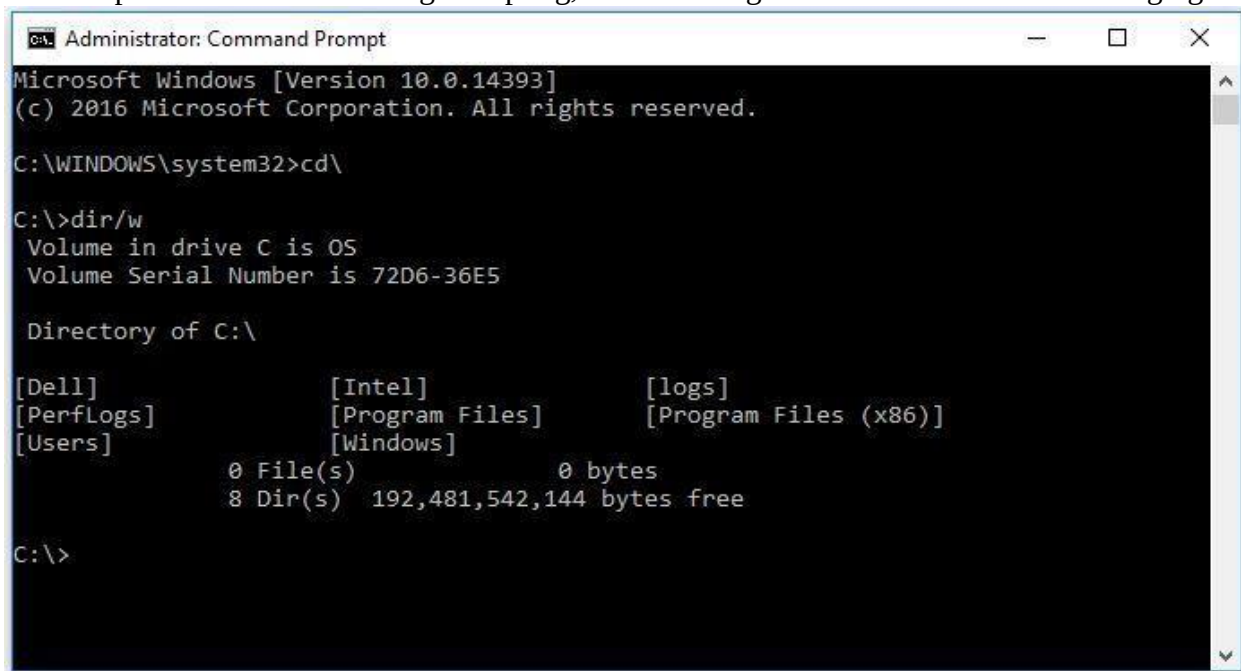
Command Prompt:

A command prompt is the input field in a text-based user interface screen for an operating system (OS) or program. The prompt is designed to elicit an action. The command prompt consists of a brief text string followed by a blinking cursor, which is where the user types command prompt commands.

Command-line interfaces (CLI) and prompts were the standard interface for computers from the early days of computing into the 1980s. Microsoft MS-DOS systems and other early consumer-based computers used CLIs. Current Windows systems offer the CLI for administrative tasks. The CLI is still an essential part of the Linux OS.

The command prompt is an executable CLI program, cmd.exe. At the command prompt, the user types a statement including a base batch file or a command name and any arguments to specify running conditions, logging and so on for the program. In Windows systems, such as Windows 10 and many previous versions of Windows, the command interpreter and executioner are referred to as the Windows Command Processor.

Command prompt interfaces can be powerful and succinct. Some tools that aren't available through the graphical user interface (GUI) can be accessed through the command prompt. It also offers superior automation through scripting, but mastering the commands can be challenging.



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.14393]
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C:\WINDOWS\system32>cd\

C:\>dir/w
Volume in drive C is OS
Volume Serial Number is 72D6-36E5

Directory of C:\

[Dell]                [Intel]                [logs]
[PerfLogs]            [Program Files]        [Program Files (x86)]
[Users]               [Windows]
                   0 File(s)                0 bytes
                   8 Dir(s)  192,481,542,144 bytes free

C:\>
```

Examples of commands

Commonly used commands include the following:

- **Cls** to clear the command prompt screen;
- **Copy** to create a copy of something such as a file;
- **Del** to delete a file;
- **Format** to formats a disk according to specific requirements;
- **Ipconfig** to list the network data such as IP address and subnet mask;
- **Ping** to send a packet to an IP address and display the information returned from the other device;
- **Sfc** to detect and fix corrupt system files; and
- Shutdown to execute a system shutdown or restart.

DOS Commands are divided into 2 types:

- **Internal Commands in Dos:** internal commands of dos are for performing basic operations on files and directories and they do not need any external file support. These DOS commands are called internal commands of dos because their code is included in COMMAND.COM, the DOS and Windows 9x command processor. Since COMMAND.COM is loaded into primary memory when DOS or Windows 9x is booted, the internal commands of dos are always in memory and can be executed at any time without first being fetched from disk.
- **External Commands of Dos :** These external commands are for performing advanced tasks and they do need some external file support as they are not stored in COMMAND.COM

Most Commonly Used Internal DOS Commands

1. DATE

This command is used to display the system current date setting and prompt you to enter a new date. The syntax is: **DATE [/T | date]**

```
C:\>date
The current date is: Tue 03/29/2011
Enter the new date: <mm-dd-yy>

C:\>date/t
Tue 03/29/2011

C:\>
```

If you type DATE without parameters then it displays current date and prompts to enter new date. We should give new date in mm-dd-yy format. If you want to keep the same date just Press ENTER. DATE command with /T switch tells the command to just output the current system date, without prompting for a new date.

2. TIME

This command is used to displays or set the system time.

The syntax is: **TIME** [/T | time]

```
C:\>time
The current time is: 13:41:53.99
Enter the new time:

C:\>time/t
01:42 PM

C:\>
```

Same as DATE command, typing TIME with no parameters displays the current time and a prompt for a new one. Press ENTER to keep the same time. TIME command used with /T switch tells the command to just output the current system time, without prompting for a new time.

3. COPY CON

It is used to create a file in the existing directory. Here CON is a DOS reserved word which stands for console.

Syntax is: **COPY CON filename** after that press Enter and start typing your text and after you're done typing your text, to save and exit hit F6 key.

4. TYPE

This command is used to display the contents of a text file or files. The syntax is: **TYPE** [drive:][path]filename

Now, lets try to display the contents of the file named filename we've created earlier using COPY CON command.

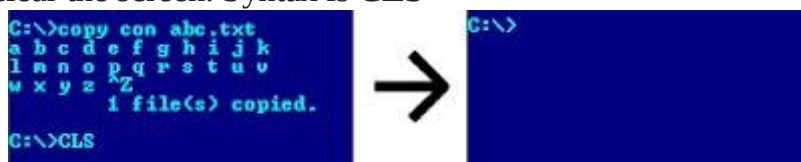
```
C:\>copy con filename
-----
COPY CON Command is used to
create a file in the existing
directory, so here we're creating
a file named filename in C drive.

To save and exit type F6, which I'm
gonna do now ^Z
      1 file(s) copied.

C:\>
```

5. CLS

It is used to clear the screen. Syntax is **CLS**



6. REN

This command is used to change/modify the name of a file or files.

Syntax is: **REN [drive:] [path] filename1 filename2.**

Here, filename1 is source file for which you wanted to change the name, and filename2 will obviously becomes your new file name. Also note that you cannot specify a new drive or path for your destination file.

7. DIR

This command displays a list of files and subdirectories in a directory. Syntax is: **DIR [drive:] [path] [filename]**

[/A[:attributes]] [/B] [/C] [/D] [/L] [/N] [/O[:sortorder]] [/P] [/Q] [/S] [/T[:timefield]] [/W] [/X] [/4]

Most Commonly Used External DOS Commands

1. EDIT

This command is used to modify or change the data of a file.

Syntax is **EDIT [/B] [/H] [/R] [/S] [filename(s)]**

Using switch /B you can force the edit in monochrome mode. /H displays the maximum number of lines possible for your system hardware. Whereas using /R and /S one can load files in read-only mode and force the use of short filenames respectively. [filename(s)] is used to specify file(s) to go edit. You can use wildcards (* and ?) to specify multiple files.

2. XCOPY

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This command is used to copy files and directory trees from one disk to another disk.

Syntax is **XCOPY source [destination] [/A | /M] [/D[:date]] [/P] [/S [/E]] [/V] [/W] [/C] [/I] [/Q] [/F] [/L] [/G] [/H] [/R] [/T] [/U] [/K] [/N] [/O] [/X] [/Y] [/Y] [/Z] [/EXCLUDE:file1[+file2][+file3]...]**

3. LABEL

It is used to create, change, or delete the volume label of a disk.

Syntax is **LABEL [drive:] [label]**

LABEL [/MP] [volume] [label]

Here, [drive:] is for specifying the drive letter of a drive to be labelled and [label] specifies the label of the volume disk. [/MP] is used to specify that the volume should be created as a mount point and [volume] is used to specify volume name, usually mentioned after drive letter followed by colon and then giving volume name required.

4. DISKCOPY

This command copies the contents of one floppy from the source drive to a formatted or un-formatted floppy disk in the destination drive. This command copies the data from particular position on the source disk to exactly the same position on the destination disk. Syntax **Diskcopy A: B:** copies contents of A: to B: drive. This command can be used with /V switch which verifies that the disk is copied correctly.

5. CHKDSK

This command is used to check a disk and display a status report with properties of disk like serial number, volume label, memory and other properties along with errors on the disk if any.

Syntax is **CHKDSK [volume**

path] [/F] [/V] [/R] [/X] [/I] [/C] [/L[:size]]

[volume path] is where you specify the drive letter followed by a colon and volume name to be checked. using /F switch allows you to fix errors on the disk. /V display full path and/or cleanup message if any. /R is used in tandem with /F and used to locate bad sectors and recover readable information. If you wanted to perform a less vigorous check of index entries on the disk then the right option is to use /I or /C rather than /R as they skip checking of cycles on the volume and helps in reducing the amount of time required to run chkdsk. Using /X forces the volume to dismount first before checking is performed. /L:size is all about specifying the log file size in kilobytes.

6. TREE

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This command is very useful to view the list of directories and subdirectories present on the disk in graphical form. If you wanted to include files also with directories and subdirectories, then you'll have to give the command line as tree/f which presents the tree view of all the content on your disk. Here is the syntax for this command with allowed switches:

TREE [drive:path] [/F] [/A]

In case you wanted use ASCII instead of extended characters, then go ahead include /A in the command line.

Task:

- Find 10 internal and external commands and implement the command.

Internal DOS Commands:

1. DIR (Directory Listing):

Purpose: List files and folders

```
09/11/2026 12:27 PM <DIR> .
09/11/2026 12:27 PM <DIR> ..
02/24/2026 03:45 PM <DIR> .vscode
03/10/2026 12:06 AM <DIR> 3D Objects
03/10/2026 12:06 AM <DIR> Contacts
06/16/2026 01:22 PM <DIR> Desktop
06/11/2026 04:55 PM <DIR> Documents
09/18/2026 11:47 AM <DIR> Downloads
03/10/2026 12:06 AM <DIR> Favorites
03/10/2026 12:06 AM <DIR> Links
03/10/2026 12:06 AM <DIR> Music
10/25/2025 04:38 PM <DIR> OneDrive
04/29/2026 04:44 PM <DIR> Pictures
03/10/2026 12:06 AM <DIR> Saved Games
03/10/2026 12:06 AM <DIR> Searches
03/10/2026 12:06 AM <DIR> Videos
          0 File(s)          0 bytes
        16 Dir(s) 54,638,448,640 bytes free
Danish>
```

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2. CD (Change working directory):

```
Danish>cd  
C:\Users\Student  
Danish>
```

3. MD (Create a new directory):

```
Danish>md MyFolder  
Danish>
```

4. RD (Remove an empty directory)

```
Danish>rd MyFolder  
Danish>
```

5. COPY (Copy file(s) to another location)

```
Danish>echo Hello World > file1.txt  
  
Danish>copy file1.txt backup.tx  
1 file(s) copied.  
  
Danish>
```

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External DOS Commands:

1. XCOPY (Copy files/directory trees recursively):

```
Danish>md C:\Source

Danish>echo Lab test > C:\Source\sample.txt

Danish>md C:\Dest

Danish>xcopy C:\Source C:\Dest /S /Y
C:\Source\sample.txt
1 File(s) copied

Danish>
```

2. ATTRIB (View or modify file attributes (R, H, A, S)):

```
Danish>ATTRIB
A             C:\Users\Student\backup.tx
A             C:\Users\Student\file1.txt
A      H      I      C:\Users\Student\NTUSER.DAT
A      SH          C:\Users\Student\ntuser.dat.LOG1
A      SH          C:\Users\Student\ntuser.dat.LOG2
A      SH          C:\Users\Student\NTUSER.DAT{7f3fd8
A      SH          C:\Users\Student\NTUSER.DAT{7f3fd8
A      SH          C:\Users\Student\NTUSER.DAT{7f3fd8
A      SH          C:\Users\Student\NTUSER.DAT{7f3fd8
A      SH          C:\Users\Student\NTUSER.DAT{7f3fd8
A      SH          C:\Users\Student\NTUSER.DAT{7f3fd8
A      SH          C:\Users\Student\ntuser.ini

Danish>
```

3. FC (Compare two files and show diffs)

```
Danish>fc file1.txt file2.txt
Comparing files file1.txt and FILE2.TXT
***** file1.txt
Hello World
***** FILE2.TXT
Lab test
*****

Danish>fc file1.txt backup.txt
Comparing files file1.txt and BACKUP.TXT
FC: cannot open BACKUP.TXT - No such file or folder

Danish>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=18ms TTL=114
Reply from 8.8.8.8: bytes=32 time=17ms TTL=114
Reply from 8.8.8.8: bytes=32 time=18ms TTL=114
Reply from 8.8.8.8: bytes=32 time=17ms TTL=114

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 17ms, Maximum = 18ms, Average = 17ms

Danish>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 4:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::cc6a:9b69:e692:a4ef%15
    IPv4 Address. . . . . : 172.16.2.201
    Subnet Mask . . . . . : 255.255.0.0
```


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4. PING (Test network/IP reachability):

```
Danish>PING

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
          [-r count] [-s count] [[-j host-list] | [-k host-list]]
          [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
          [-4] [-6] target_name

Options:
  -t          Ping the specified host until stopped.
              To see statistics and continue - type Control-Break;
              To stop - type Control-C.
  -a          Resolve addresses to hostnames.
  -n count    Number of echo requests to send.
  -l size     Send buffer size.
  -f          Set Don't Fragment flag in packet (IPv4-only).
  -i TTL      Time To Live.
  -v TOS      Type Of Service (IPv4-only. This setting has been deprecated
              and has no effect on the type of service field in the IP
              Header).
  -r count    Record route for count hops (IPv4-only).
  -s count    Timestamp for count hops (IPv4-only).
  -j host-list Loose source route along host-list (IPv4-only).
  -k host-list Strict source route along host-list (IPv4-only).
  -w timeout  Timeout in milliseconds to wait for each reply.
  -R          Use routing header to test reverse route also (IPv6-only).
              Per RFC 5095 the use of this routing header has been
              deprecated. Some systems may drop echo requests if
              this header is used.
  -S srcaddr  Source address to use.
  -c compartment Routing compartment identifier.
  -p          Ping a Hyper-V Network Virtualization provider address.
  -4          Force using IPv4.
  -6          Force using IPv6.
```

5. SFC (Scan and repair protected system files)

```
Danish>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=42ms TTL=112
Reply from 8.8.8.8: bytes=32 time=48ms TTL=112
Reply from 8.8.8.8: bytes=32 time=47ms TTL=112
Reply from 8.8.8.8: bytes=32 time=26ms TTL=112

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 26ms, Maximum = 48ms, Average = 39ms

Danish>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 4:

    Connection-specific DNS Suffix . . . : 
    Link-local IPv6 Address . . . . . : fe80::cc6a:9b69:e692:a4ef%15
    IPv4 Address. . . . . : 172.16.2.201
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : 172.16.0.9
```

Conclusion:

This lab demonstrated practical command-line proficiency in cmd.exe via custom prompt configuration (Danish>), directory navigation, and file manipulation. Through hands-on execution, distinction was established between **Internal commands** (DIR, CD, MD, RD, COPY, TYPE, CLS) executing natively within the shell and **External/Utility commands** (FC, PING, IPCONFIG) requiring dedicated binaries or specific parameters. Troubleshooting path and privilege errors reinforced critical OS navigation and administration fundamentals.